

work was implemented as a Linux device driver for a commercial underwater network modem SM-75 by Teledyne Benthos. It was tested by dropping two 18.1kg (40-pound) sensors into Lake Erie, near Buffalo, and communicating with them wirelessly using a conventional laptop. This test is a precursor, showing the possibility of implementing the technology in deep waters too.

Sensors and other devices connected using this technology, which they hope to develop into a standard, would be able to readily communicate with other commercial, terrestrial wireless networks—relevant information will be available to any authorised person with a smartphone or computer connected to the Internet! So, if a sensor network detects a tsunami, warnings can be directly issued to all concerned officials or even the public. Sea animals can be monitored. Sly submarines and drug trafficking can be easily prevented. A lot more can be done, basically because the proposed undersea wireless network would be able to connect to the Internet!

Outer space Internet

When we see something new, we want to say lots about it! That is human tendency. Imagine how much an astronaut would want to communicate about the totally unexplored frontiers he beholds. Radio-based communications, of course, is possible from outer space, but NASA felt that it was not efficient enough to transmit heavy videos and data loads from spacecrafts, and so they set about exploring laser-based optical communications. Around a year ago, NASA set about testing a high data-rate laser communication system aboard its lunar spacecraft, the Lunar Atmosphere and Dust Environment Explorer (LADEE). The tests were phenomenally successful, opening the possibility of replacing radio systems with laser systems for faster satellite communications and deep space communications with human or robotic exploration teams.

During the tests, NASA observed that a laser beam sent from an Earth station to LADEE could relay data at the rate of 20Mbps. The downlink, from the

More deep sea talks

► Quantum key distribution makes top-secret communications possible, and for a long time now naval officers have been interested in knowing if it is possible to implement it underwater to communicate with submarines. Recently, a team at the Ocean University of China in Qingdao calculated how far photons could travel in clear ocean water without damaging the information in them. It concluded that quantum-secured content can be sent at a rate of around 215 kilobits per second to distances within 125 metres under water.

► Dr Christian Schlegel of Dalhousie University recently received more than ₹ 140 million (\$2.3 million) funding for his research into new algorithms that support high-speed underwater communications networks. Acoustics is the main medium for underwater communications today, but it has to contend with the high noise levels in subsea channels. Dr Schlegel and his team aim to develop industry-usable computational algorithms that deal with this specific problem.

► Thinking beyond sound and radio waves, Professor Zhaowei Liu with others at the University of California, San Diego, is working towards developing an underwater optical communication system using fast-blinking LED systems. At the heart of the proposed system is a new artificial meta-material that can increase the light intensity and blink speed of a fluorescent light-emitting dye molecule.

spacecraft to the Earth station, could relay data at the rate of 622Mbps. The concerned team noted that six-times-faster communication from the moon was possible with laser instruments that were just half the mass, half the weight and used 25 per cent lesser power compared to radio equipment. Implementing this system in future spacecrafts would make it possible to explore Mercury, Mars and much more of outer space by relaying HD videos captured there, or even telepresence that allows scientists to virtually be there!

Close on the heels of this phenomenal success, NASA is expected to launch the Laser Communications Relay Demonstration (LCRD; <http://esc.gsfc.nasa.gov/267/LCRD.html>) sometime in 2017, which will test the success of this communication system over a multi-year trial period.

Scaling great heights

Humans seem to be finding some means to communicate with each other from the nooks and corners of the Earth. While at one point of time, mountaineers were totally cut off from base after a certain altitude, today they can communicate even from the middle of nowhere with satellite phones and satellite constellations like Iridium.

In 2012, 46-year old Simone Moro, master of over 50 mountaineering expeditions, set his mind on conquering Nanga Parbat, known in trekking circles as the 'Killer Mountain' under harsh Himalayan winter conditions. Till then, this peak had been impossible to reach

in the winters because the difference in height between its summit and the nearest base camp is the largest in the world, and it is notorious for its unpredictable snow storms. What made Moro think he could do it?

Since 12 years before that, he had been using the Thuraya mobile satellite communications equipment with support from Intermatica, and with its help he became the first alpinist to conquer Mount Everest with a satellite phone. He was confident he could reach up Nanga too, with the help of a similar communication system.

His hunch worked. On the journey to Nanga, Moro records having escaped a killer storm. When he contacted the weather forecast centre in Austria with his satellite phone, he was alerted of the incoming storm, which helped him and his partner to outsmart the invincible downpour of snow, and ultimately reach the summit of Nanga Parbat.

For a long time now, satellite phones have been used by defence organisations for coordinating with personnel posted in remote areas. In India too, satellite communication is used mainly by the defence forces. In fact, there is such a great demand on this front that the Telecom Regulatory Authority of India (TRAI) has been pushing the need for Bharat Sanchar Nigam Limited (BSNL) to set up a new gateway for satellite phone services, which will address the rising requirement from security forces.

It is also interesting to note that in September this year, the media reported about the Indian government's intent to